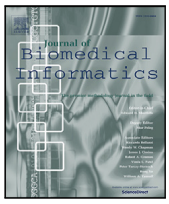
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Original Research



Patient deep spatio-temporal encoding and medication substructure mapping for safe medication recommendation

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ABSTRACT

Medication recommendations are designed to provide physicians and patients with personalized, accurate and safe medication choices to maximize patient outcomes. Although significant progress has been made in related research, three major challenges remain: inadequate modeling of patients' multidimensional and time-series information, insufficient representation of medication substructures, and poor balance between model accuracy and drug-drug interactions. To address these issues , a safe medication recommendation model SDRBT based on patient deep spatio-temporal encoding and medication substructure mapping is proposed in this paper. SDRBT has developed a patient deep temporal and spatial coding module, which combines symptom information, disease diagnosis information, and treatment information from the patient's electronic health record data. It innovatively utilizes the Block Recurrent Transformer to model longitudinal temporal information of patients in different dimensions to obtain the horizontal representation of the patient's current visit. A dual-domain mapping module for medication substructures is designed to perform global and local mapping of medications, fully learning and aggregating medication substructure representations. Finally, a PID LOSS control unit was designed, in which we studied a drug interaction control module based on the similarity calculation between the electronic health map and the drug interaction graph. This module ensures the safety of the recommended medication combination effectively improved the recommendation efficiency and reduced the model training time. Experiments on the public MIMIC-III dataset demonstrate SDRBT's superior accuracy in medication recommendation.

1. Introduction

Medication recommendation, as an indispensable part of the modern healthcare system, is gradually becoming a key means to improve patient treatment outcomes and optimize the allocation of medical resources [1–4]. Currently, there are three mainstream approaches for medication recommendation: The first method is rule-based recommendation [5,6], which relies on content from medical expert knowledge bases. The second method is instance-based recommendation [7,8], which typically focuses on the patient's current health status and extracts feature information from the patient's current medical records for medication recommendation. The third method is longitudinal information-based recommendation [9–11], which significantly enhances the accuracy and personalization level of medication recommendation by mining relevant information from patients' historical medical records. Although significant progress has been made in longitudinal information-based medication recommendation, it still faces the following challenges:

Question 1: How to fully model the longitudinal information in EHR data to better utilize relevant information from different dimensions [12,13]? Fig. 1 shows the number of patients with different visit counts in the MIMIC-III dataset. Our analysis reveals that only 5.39% of patients have a single visit record, while the vast majority of patients in the EHR data have multiple visit records at different times. Currently, most mainstream research has primarily used diagnostic and procedural information of patients, typically modeling these with Transformer models. However, Transformers do not perform well in modeling long-term time series tasks, which may lead to certain limitations in the resulting patient-level representations [14].

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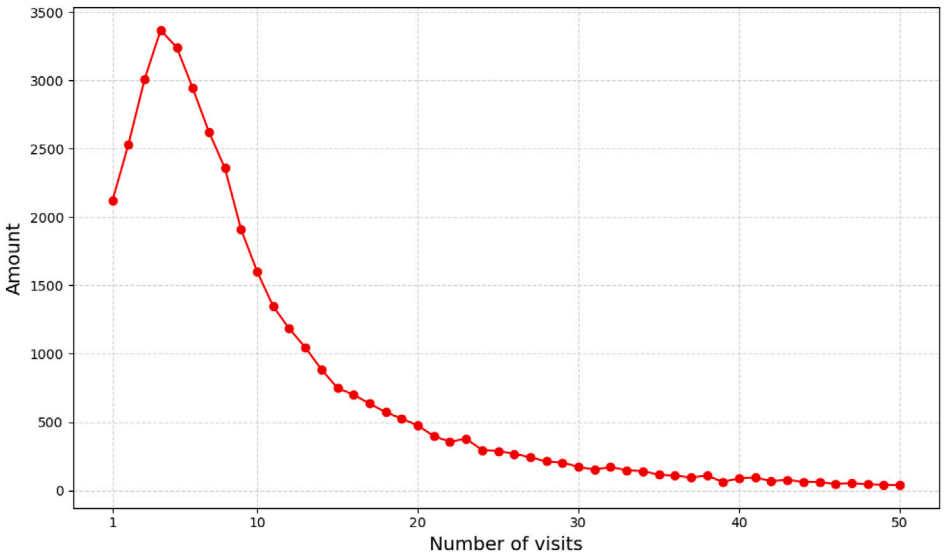


Fig. 1. The number of patients with different visit counts in the MIMIC-III dataset.

Question 2: How to adequately represent the molecular structure of s and identify substructures that play a crucial role in treating patients' diseases [8,15]? Each medication is composed of various atoms and atomic groups connected through chemical bonds, and it is the functional groups of medications that truly play a significant role in the treatment of patients' diseases [16]. Current work mainly focuses on recommending medications for the current visit based on patients' past medication history, neglecting the important role of medication molecular structure in medication recommendation [17].

Question 3: How to enhance the model's recommendation accuracy while reducing drug-drug interactions (DDI) [3,18]? Drug-drug interactions refer to specific interactions between drugs that can enhance or weaken their efficacy or even trigger adverse reactions [19].

To address the aforementioned issues, this paper proposes a safe medication recommendation model SDRBT based on medication substructure mapping and deep spatio-temporal coding. SDRBT combines symptom information, disease diagnosis, and treatment information from the patient's EHR data, using Block Recurrent Transformer (BRT) [20] to model longitudinal EHR data. Block Recurrent Transformer integrates the recursive mechanism of LSTM with the cyclical mechanism of RNN, significantly improving the ability to model long-term sequences. By introducing BRT, SDRBT is capable of fully modeling the sequential visit information of the same patient across different times in the EHR data, comprehensively considering each patient's longitudinal information, thus addressing the limitations of current visit-based representations. Moreover, to fully represent the molecular substructures of medications, we designed a dual-domain mapping module. This module performs global and local dual mappings of medication molecules, enabling the identification of medication substructures that play a significant role in the treatment of the patient's disease. We then use the proportion of medication substructures across different medications to recommend medications based on the patient's current visit information, further improving the accuracy of the medication package recommendation. Finally, the model integrates the EHR graph with the DDI graph, performing similarity calculations to select medications used during the patient's treatment and supplement drug interactions missing from the DDI graph. This integration effectively reduces the model's training time and improves its efficiency. At the same time, we designed a PID Loss control unit to account for drug-drug interactions. The model incorporates the DDI control module, which is derived from similarity calculations, into the PID Loss control unit. During training, it intervenes in the probability of each medication appearing in the treatment of the patient's disease, ensuring the safety of the medication combination. Overall, our main contributions are mainly in the following four areas:

- Using Block Recurrent Transformer to model longitudinal patient information and related multi-dimensional data. The model combines symptom information with diagnostic and procedural data to obtain patient-level representations, fully utilizing longitudinal information and combining multi-dimensional data.
- Designing a medication substructure dual-domain mapping module, which introduces an attention learning method based on medication molecular substructures. This module effectively learns and aggregates medication substructure representations. By integrating global medication representations and medication substructure representations, the model achieves a substructure-aware representation for final medication recommendations.
- Integrating EHR graphs with DDI graphs, performing similarity calculations to construct the DDI control module, and incorporating it into the PID Loss control unit. This integration enhances model efficiency while ensuring fine-tuned control over the safety of medication combinations.
- Conducting experiments on the publicly available MIMIC-III dataset [21] to demonstrate the superiority of SDRBT, and performing ablation studies to validate the effectiveness of the model components. The experimental results show that the SDRBT model outperforms other baseline models in recommendation performance, providing more accurate medication package recommendations with lower DDI risks.

2. Related works

2.1. Medication recommendations

Medication recommendation can be broadly categorized into three main types from its emergence and development to date: rule-based medication recommendation, instance-based medication recommendation, and longitudinal information-based medication recommendation.

(1) Rule-based medication recommendation mainly relies on a series of predefined rules or algorithms to recommend medications for patients or healthcare professionals [6,22]. The disadvantage of this approach is that rules are often based on known and limited datasets and situations also leads to the fact that such methods are not flexible enough to be adapted to new or complex cases.

(2) Instance-based medication recommendation recommends medications based on the patient's current visit information [3]. LEAP [8] uses an innovative multi-instance multi-label learning framework to

decompose medication recommendations into a sequential decision-making process. The framework employs RNN and attention mechanisms to deeply analyze and encode the patient's current visit, and also introduces reinforcement learning to prevent undesirable medications from being generated, which improves the accuracy of the medication recommendation. However, this type of approach mainly relies on the patient's current diagnosis and procedure records when dealing with medication recommendation, ignoring the patient's rich historical medical history information, which may affect the accuracy and safety of the recommendation results.

(3) Medication recommendation based on longitudinal information considers [15,23] not only the patient's current visit information, but also longitudinal information such as the patient's past medical history and treatment experience. SafeDrug [12] is the first model to model medication molecules, which is equipped with a global message-passing neural network module and a local dichotomous learning module to achieve complete encoding of medication molecule connectivity and functionality; COGNet [24] proposes a replication or prediction mechanism, which further transforms the patient medication recommendation task into a patient medication change prediction problem, modeling the relationship between medication changes during consecutive patient visits, thus further improving the performance of medication recommendation.

2.2. Molecular graphical representation

Representing medication molecules in the form of graphs can capture the complex internal structures and interactions of molecules, which improves the accuracy of predicting medication properties and functions, thereby enabling more precise medication recommendations. Graph Neural Networks (GNNs) have demonstrated significant capabilities in handling graph-structured data [25,26]. They iteratively update node features using neighbor aggregation strategies and perform classification or regression on the nodes in the graph. Due to the relationships between medication molecular structures and properties, such as the effectiveness and safety of medication molecules, some studies have attempted to use GNNs to enhance the performance of medication recommendation models. MPNN [27] is a GNN method based on a message-passing mechanism that updates atomic embedding vectors by exchanging information between atoms and bonds; D-GCAN [28] uses graph convolution and graph attention to perform molecular representation and importance calculation, forming molecular representations; Shang et al. [15] employed GNNs to encode Electronic Health Records (EHR) and Drug-Drug Interaction (DDI) graphs to introduce medication-related information; Zhang et al. [8] segmented medications based on functional groups and performed graph encoding on each segmented part to comprehensively describe the function and structure of the medication.

3. Problem formulation

In the medication recommendation task, we use the patient's EHR information to provide personalized medication combination recommendations for the patient. In this section, we present the EHR data, relevant medical information, and the goal of the medication recommendation problem.

3.1. Electronic Health Record(EHR)

Electronic Health Records (EHR) consist of visit information from multiple patients [29], which detailedly record patients' vital signs, examination results, diagnosis outcomes, treatment processes, medication usage, and other information. The longitudinal cases of all patients can be represented as $V = \{V^{(1)}, V^{(2)}, \ldots, V^{(N)}\}$, where N is the total number of patients.

For a specific patient i, their electronic health record can be viewed as a multivariate visit sequence composed of n visit records from patient i: $V^{(i)} = \{v_1^{(i)}, v_2^{(i)}, \ldots, v_n^{(i)}\}$. Specifically, the tth visit information of patient i can be further represented as $V_t^{(i)} = \{s_t^{(i)}, d_t^{(i)}, p_t^{(i)}, m_t^{(i)}\}$, where $s_t^{(i)} \in \{0,1\}^{|S|}$ is the symptom vector, $d_t^{(i)} \in \{0,1\}^{|D|}$ is the diagnosis vector, $p_t^{(i)} \in \{0,1\}^{|P|}$ is the procedure vector, and $m_t^{(i)} \in \{0,1\}^{|M|}$ is the medication vector. S, D, P and M respectively represent the overall symptom set, diagnosis set, procedure set, and medication set.

3.2. EHR and DDI charts

For ease of understanding, we can formalize the EHR graph and DDI graph as $G_e = \{M, X_e\}$ and $G_d = \{M, X_d\}$ respectively. Here, M represents the set of all medications, X_e represents the known medication combinations in the EHR graph, and X_d represents the known drug interaction combinations. We define adjacency matrices $E_e, E_d \in R^{|M*M|}$ to help elucidate the construction of the edge sets X_e and X_d. For matrix E_e, if medication i is present in the jth medication combination recommendation, then E_e[i, j] = 1; for matrix E_d, if there is an interaction between medication i and medication j, then E_d[i, j] = 1.

3.3. Medication recommendation issues

Given the longitudinal electronic health record information of a patient, the EHR graph, and the DDI graph, the task objective of medication recommendation is to provide a suitable, targeted, and relatively safe combination of medication packages, represented as vector y, for the patient's current visit based on their known medical visit information. Here, $y \in \{0,1\}^{1*|M|}$. In this context, M represents the set of all medications. If the value of medication A in y is 1, it indicates that the medication is recommended; if the value is 0, it indicates that the medication is not recommended.

4. Modeling framework

In this section, we will introduce the technical details of the SDRBT model in depth. As shown in Fig. 2, SDRBT consists of four main modules: (a) Multi-dimensional information aggregation module, which models disease diagnosis information, treatment information, and patient symptom information in the patient's EHR data, preprocesses the patient's visit information, and obtains the output representation of the patient's different dimensional information at time t; (b) Longitudinal characterization aggregation module, which uses Block Recurrent Transformer instead of Transformer to fully model the patient's time-series visit information at different times, comprehensively considers the longitudinal information of each patient, and obtains a relatively complete representation of the patient's current visit information; (c) Medication structure dual domain mapping module, in which we perform global and local dual mapping of medication molecules, fully represent the molecular substructures of medications, learn the correlation between medication substructures and patient health, and thereby screen out medication substructures that play an important role in the treatment of patient diseases; (d) Safe medication combination recommendation module, in which we design a PID Loss control unit to control medication interactions. We integrate the EHR graph and DDI graph and perform similarity calculations, incorporate the similarity-calculated DDI control module into the PID Loss control unit, and intervene in the probability of each medication appearing in the treatment process during training to ensure the safety of medication combinations.

4.1. Multi-dimensional information aggregation module

In the multidimensional information aggregation module, in addition to following the traditional patient data processing methods,

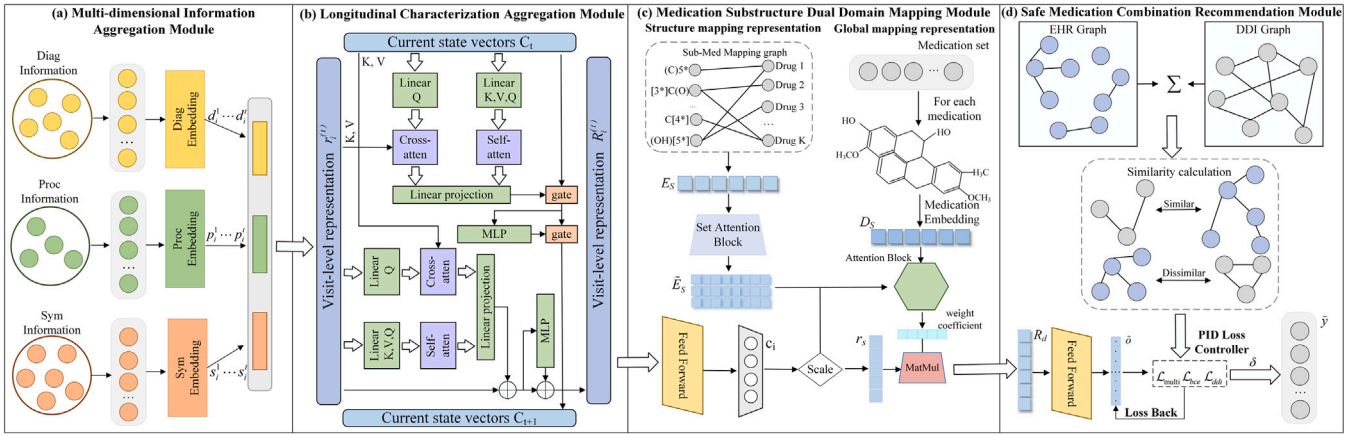


Fig. 2. The architecture of model SDRBT: (a) Multi-dimensional information aggregation module; (b) Longitudinal characterization aggregation module; (c) Medication structure dual domain mapping module; (d) Safe medication combination recommendation module.

converts the ICD code into the corresponding text description in the dictionary table, to extract the text content for diagnosis and procedure, and combines the patient symptom description information extracted from the clinical record. Subsequently, we encode the disease diagnosis information, symptom description information, and procedure record information from patient i's historical t medical records. The representations of these three types of information after encoding are $d_i^1, d_i^2 \cdots d_i^t$, $s_i^1, s_i^2 \cdots s_i^t$, and $p_i^1, p_i^2 \cdots p_i^t$. Finally, we concatenate these three types of information to form a patient visit-level representation, which serves as the patient's health status during the current visit.

The visit-level representation $r_i^{(t)}$ is defined as:

$$r_i^{(t)} = \left[d_i^1, d_i^2 \cdots d_i^t \| s_i^1, s_i^2 \cdots s_i^t \| p_i^1, p_i^2 \cdots p_i^t\right] \tag{1}$$

4.2. Longitudinal characterization aggregation module

In the longitudinal characterization aggregation module, we use Block Recurrent Transformer instead of the traditional Transformer to process the patient's past t visits' information, thereby obtaining the current visit representation of the patient. The state vector C_t carries the patient's temporal information, enabling the model to understand changes in the patient's state over time. The visit-level representation $r^{(t)}$ updates information based on historical state representations. The update function for the state vector C_i is represented as follows:

$$z = \tanh(W_z Y + b_z) \tag{2}$$

$$i = \sigma(W_i Y + b_i - 1) \tag{3}$$

$$f = \sigma(W_f Y + b_f + 1) \tag{4}$$

$$g(X, Y) = X \odot f + z \odot i \tag{5}$$

$$C_t' = Y_c'([Att(C_t, C_t, C_t), Att(C_t, r^{(t)}, r^{(t)})]) + q_c' \tag{6}$$

$$C_{t+1} = g(MLP(g(C_t', C_t)), g(C_t', C_t)) \tag{7}$$

In the formula, trainable weight matrices W_z, W_i, W_f and trainable bias vectors b_z, b_i, b_f are introduced. C_t' is the combination of masked self-attention on the current hidden state C_t and masked cross-attention with the access-level representation $r^{(t)}$. Y_c' and q_c' are two learnable parameters.

$r^{(t)\prime}$ is the concatenation of masked self-attention and masked cross-attention with the current state in the access-level representation. Its core characteristic is delegating a significant portion of the responsibility for information updating to the generated attention weights. The representation of $r^{(t)\prime}$ is given by the following formula, where Y_v' and q_v' are two learnable parameters.

$$r^{(t)\prime} = Y_v'([Att(r^{(t)}, r^{(t)}, r^{(t)}), Att(r^{(t)}, C_t, C_t)]) + q_v' \tag{8}$$

The update flow for the current visit representation $R^{(t)}$ selects longitudinal information from the hidden state and visit-level information from the current visit. The representations obtained above are concatenated, and the MLP technique is used to obtain the patient's current visit representation $R^{(t)}$ as follows:

$$R^{(t)} = MLP(r^{(t)\prime} + r^{(t)}) + (r^{(t)\prime} + r^{(t)}) \tag{9}$$

The patient deep spatio-temporal coding consists of two modules: the multi-dimensional information aggregation module, which is responsible for encoding spatial information at different levels of the patient, and the longitudinal characterization aggregation module, which is responsible for encoding temporal information of the patient. After passing through these two modules, the patient's EHR data can be encoded into the representation of the patient's current visit, which is used for processing in subsequent modules.

4.3. Medication structure dual domain mapping module

In the medication structure dual domain mapping module, we perform global and local dual mapping for each type of medication appearing in the patient's EHR data, thereby fully representing the substructural features of medication molecules and effectively learning and aggregating the representations of these substructures. By integrating the relevant content of global medication representations and medication substructure representations, we can analyze the correlation between medication substructures and patient diseases, obtain their contribution to disease treatment, and screen out the substructures that play a key role in the treatment of patient diseases.

In the process of global medication mapping, we use the Graph Isomorphism Network (GIN) [30] to encode the medications appearing in the patient's EHR data. GIN is capable of iteratively updating node features using a neighbor aggregation strategy, as shown in formula (10). We define an embedding table D_S to store the global mapping representations of all medications. Through a function $L_y(Z)$, we extract the P-layer that aggregates all node features and store it in D_S, where each row corresponds to a specific medication.

$$h_v^{(k)} = MLP^{(k)}((1 + \varepsilon^{(k)}) \cdot h_v^{(k-1)} + \sum_{u \in N(v)} h_u^{(k-1)}) \tag{10}$$

$$L_y(Z) = READOUT(\{h_P^{(k)}(u)|u \in V\}) \tag{11}$$

Each medication is composed of multiple specific molecular substructures, and it is the functional groups of the medication that truly play a crucial role in the treatment of patients' diseases. In the process of local medication mapping, we decompose the medications appearing in the patient's EHR data into substructures based on the molecule-medication mapping diagram. An embedding table E_s is defined to store the decomposed medication substructures, where each row represents the embedding of a specific substructure.

Subsequently, we use the Set Attention Block (SAB) [31] as the substructure interaction module to model the substructures. The reason for using SAB to handle substructure interactions is that it can effectively capture high-order interactions between different substructures. Medications are typically composed of multiple substructures, and these substructures do not function in isolation during treatment; rather, they interact with each other. Through the self-attention mechanism, SAB can model the relationships between these substructures without considering their sequence, thereby capturing the high-order interaction information between them. This enables us to gain a more comprehensive understanding of each substructure's contribution to the medication's therapeutic effect, learn finer-grained features of the medication, and provide support for recommending more clinically relevant medication combinations.

The modeling process of SAB is shown in formulas 12–13, where dk is the dimension of the key embedding, and Q_x, K_x, and V_x are the query, key, and value matrices obtained by applying different transformations to the input X, respectively. FF1 represents the feedforward network, and LN denotes layer normalization. In the SAB module, we compute the inner product of Q and K to obtain the similarity between the substructures, which determines the importance of each substructure in the model. We then perform a weighted summation of V to generate the updated substructure representations. We define an embedding table $\tilde{E}_S$ of the same size as E_s to store the pairwise interactions between substructures after SAB modeling.

$$A(X) = \text{Softmax}\left(\frac{Q_x K_x^T}{\sqrt{d_k}}\right) V_x \tag{12}$$

$$\tilde{E}_s = LN(LN(E_s + A(E_s)) + FF_1(LN(E_s + A(E_s)))) \tag{13}$$

Due to the varying degrees of importance of different substructures in a medication during the patient's disease treatment process, we aim to quantify the contribution of these substructures to the treatment process, thereby screening out the medication substructures that play a crucial role in the treatment of the patient's disease. This will enable us to utilize the proportion of medication substructures in different medications for medication recommendation in subsequent modules, further enhancing the performance of the model's recommendations.

To achieve this goal, we employ a feedforward neural network with a Sigmoid activation function to process the current patient visit representation $R_i^{(t)}$ passed into the module. Additionally, we define an embedding table c_i to record the contribution of each medication substructure to the treatment of the disease during the patient's tth visit, as shown in formula (14).

$$c_i = \sigma(FF_2(R_i)) \tag{14}$$

With the various embedding table information obtained above, given a medication molecule e, we can extract its global medication representation r_e from D_S and its local medication representation l_e from $\tilde{E}_S$. We define the total number of substructures contained in each medication as the set S_k, and each medication molecule substructure a is represented as r_a. Subsequently, we use the Attention Block to calculate the attention coefficient $\partial_{k,a}$, which indicates the importance of medication molecule substructure a to the entire medication, as shown in formula (15). In this formula, W_g and W_s correspond to the linear transformations of r_t and r_s, respectively. For each medication molecule substructure a, we can scale its representation based on its contribution to the patient's disease, as shown in formula (16).

$$\partial_{k,a} = \frac{\exp(r_e W_g W_s^T r_a^T)}{\sum_{i \in S_k} \exp(r_e W_g W_s^T r_i^T)} \tag{15}$$

$$\forall a \in S_k, r_s = c_i l_e \tag{16}$$

Finally, the scaled medication molecular substructure representations are aggregated to obtain the substructure-aware representation of a given medication molecular substructure A. The substructure aware representation of all medications constitutes the embedding table R_d, where K is the total number of medications.

$$R_d = \sum_{i \in K} \sum_{a \in S_k} \partial_{k,a} r_s^{(i)} \tag{17}$$

4.4. Safe medication combination recommendation module

The safe medication combination recommendation module can use the substructure-aware representation of medication molecules obtained in the medication structure dual-domain mapping module to provide medication recommendations for the patient's current visit. In order to obtain the recommendation probability $\tilde{o}$ of each medication in this visit record, we use a feedforward neural network with Sigmoid activation function to process the substructure perception representation table R_d of the medication, as shown in formula (18):

$$\tilde{o} = \sigma(FF_3(R_d)) \tag{18}$$

There may be certain interactions between each medication. If the medications are simply combined, the DDI rate in the final recommended medication package may be high, and there may be adverse reactions that are harmful to the human body. To solve this problem, we first design a similarity calculation block. In this block, we integrate the EHR map with the DDI map, and calculate the similarity between them, so as to screen the medications used by the current patient and supplement the drug interactions that do not appear in the DDI map, so as to improve the efficiency and accuracy of the model recommendation.

We use Graph Convolutional Neural Network (GCN) [32] to learn the features of EHR graphs and DDI graphs. The GCN process is shown in formula (19). Where $\tilde{D}$ is the diagonal matrix and I is the identity matrix.

$$GCN(A, W) = ReLU(\tilde{D}^{-\frac{1}{2}}(A + I)\tilde{D}^{-\frac{1}{2}}W) \tag{19}$$

We adopt a two-layer GCN to learn features on two graphs. The specific processing steps of EHR graph G_e and DDI graph G_d are as follows:

$$G_e' = GCN(G_e, GCN(G_e, W_e)W_1) \tag{20}$$

$$G_d' = GCN(G_d, GCN(G_d, W_d)W_2) \tag{21}$$

Then, we connect the processed EHR map G_e' and DDI map G_d', calculate the similarity between them, and store the corresponding results in the similarity calculation block T. In the formula, the Simil function calculates the similarity between the features of two graphs G_e' and G_d', realizes the medication screening for the current patient and supplements the drug interactions that do not appear in the DDI graph.

$$T = Simil(G_e', G_d') \tag{22}$$

At the same time, we designed a PID Loss control unit to consider the interaction between medications. Model adds the DDI control module after similarity calculation into the PID Loss control unit. During the training process, the probability of occurrence of each medication to treat the patient's disease is intervened to ensure the safety of the medication combination. Specifically, the unit contains three loss functions, namely, DDI loss L_{ddi} based on similarity calculation of block T, multi-label edge loss L_{multi} and binary cross entropy loss L_{bce}. The unit connects the three loss functions by a certain ratio, as shown in formula (23).

$$L_{ddi} = \sum_{i=1}^{|K|} \sum_{j=1}^{|K|} D_{ij} \cdot \tilde{o}_i^{(t)} \cdot \tilde{o}_j^{(t)} \tag{23}$$

$$L_{multi} = \sum_{i,j:m_i^{(t)}=1,m_j^{(t)}=0} \frac{\max(0, 1 - (\tilde{o}_i^{(t)} - \tilde{o}_j^{(t)}))}{|K|} \tag{24}$$

$$L_{bce} = -\sum_{i=1}^{|K|} m_i^{(t)} \log(\tilde{o}_i^{(t)}) + (1 - m_i^{(t)}) \log(\tilde{o}_i^{(t)}) \tag{25}$$

$$L = (1 - \alpha)(\lambda L_{bce} + (1 - \lambda) L_{multi}) + \alpha L_{ddi} \tag{26}$$

where K represents the total number of medications that have appeared in the patient's EHR data, and $m^{(t)}$ represents the multi-heat vector recommendation value of the target medication. During the training, we set a DDI rate threshold. Once the DDI rate for a single sample exceeds this threshold, negative DDI signals are especially highlighted and back-propagated. In addition, this adaptive gradient descent method [33] can improve the model recommendation accuracy while reducing the drug interaction, effectively balance the final DDI level.

Finally, medications whose recommendation probability is greater than threshold δ are selected from $\tilde{o}$ and combined to form the predicted medication set $\tilde{y}$.

5. Experimental design and analysis

In this section, we present the dataset used and assess the performance of the model SDRBT. In particular, the aim is to answer the following research questions:

RQ1: How does the SDRBT model compare to current well-performing medication recommendation methods?

RQ2: How do the component modules of the SDRBT model perform?

RQ3: How efficient is the SDRBT model compared to the baseline model?

5.1. Data sets

We evaluated the SDRBT model using the MIMIC-III dataset [21], which is currently the most comprehensive open-source clinical dataset. This dataset includes clinical diagnostic reports from patients admitted to the ICU between 2001 and 2012, containing patient information, prescription records, diagnosis codes, and other relevant data. Over 90% of the patients in this dataset were diagnosed with multiple diseases, which meets the task scenarios required for medication recommendation by the model. DDI knowledge was extracted from the TWOSIDES dataset, and we selected the top 40 severe DDI types for DDI rate evaluation.

To further obtain hierarchical information and facilitate the calculation of DDI scores, we converted NDC codes into uniform ATC code representations. Additionally, to reduce the label space of medications and simplify the problem, we used only the first three levels of the ATC system codes. To ensure data accuracy, we preprocessed the dataset using the longitudinal patient representation module. The statistics of the preprocessed dataset are shown in Table 1.

Table 1
Statistical data for the final data set.

Items	MIMIC-III
# of patients	5208
# of clinical visits	13 490
sypt./ diag. space size	428/1895
prod./med. space size	1378/112
avg. # of visits	2.59
avg. # of sypt./ diag. per visit	7.67/10.24
avg. # of prod./ med. per visit	3.85/11.30
total # of DDI pairs	337

5.2. Baseline model

To demonstrate the validity of the model, the SDRBT model is compared with the following 11 state-of-the-art methods.

The comparative instance-based medication recommendation model includes the following parts:

- LR (Logistic Regression) is a linear model of binary classification that classifies each medication label by a logistic regression model.

- ECC [7](2011) is a popular multi-label classification method with correlated labels, which uses multiple SVM classifiers for multi-label prediction of medications.

- LEAP [8](2017) is a typical instance-based approach that utilizes LSTM to extract valid information from patient records and make personalized medication recommendations based on current patient diagnostic information.

The comparative longitudinal-based medication recommendation model includes the following parts, which have been divided into smaller subgroups for clarity:

(1) RNN-based Methods:

- RETAIN [34](2016) utilized a two-layer recurrent neural network (RNN) to model EHR information and leverage attention mechanisms to improve the accuracy of medication predictions.

- FDIRNet [35](2024) adopts the forward data interpolation method to deal with the situation where there may be incomplete data or abnormal data in the EHR, and uses RNN and ResNet to capture the change pattern of patient history information, so as to improve recommendation performance.

(2) Memory and Graph Neural Network-based Methods:

- GAMENet [15](2019) used memory neural networks to store historical prescription information and graph convolutional networks to encode EHR and DDI maps.

- SafeDrug [12](2021) uses bimolecular encoders to encode the molecular structure of medications, and combines medication molecular maps with DDI maps to improve medication recommendation performance and reduce DDI.

(3) Prediction and Replication-based Methods:

- COGNet [24](2022) introduces a replication/prediction mechanism that complements current medication combination recommendations by replicating historical medications in prescriptions.

- StartMed [17](2024) overcomes the long-tail problem in medical data by introducing a hierarchical strategy, so as to learn sparse data more comprehensively.

(4) Molecular-based Methods:

- MoleRec [36](2023) proposed a method based on molecular substructure coding to model the association between specific molecular substructures within medication molecules and patients' health status, effectively identifying the relationship between medication substructure perception, so as to improve the effect of medication recommendation.

- ACDNet [18](2024) proposes an attention-based collaborative decision making network that can effectively capture diagnosis and medication status from patient visit information, and uses a collaborative decision framework to facilitate the medication recommendation process.

Table 2
Hyper parameter of experiment.

Hyper parameters description	Value
Learning rate of Adam	0.001
Target DDI	0.06
Embedding dimensions size	128
Parameter λ	0.95
Dropout rate	0.5
Epochs	50

5.3. Implementation details

In this section, we outline the implementation details of SDRBT. Based on previous studies [17,36], we split the dataset into training, validation, and test sets in a 4:1:1 ratio. All experiments were conducted on an Ubuntu 20.04 machine equipped with 64 CPU cores, 22 GB of RAM, and a 24 GB NVIDIA RTX 4090 GPU. Through parameter experiments, we selected the optimal hyperparameters for the model, setting the embedding dimension to 128 and λ to 0.95. We used the Adam optimizer with an initial learning rate of 0.001, training the model for 50 epochs. To prevent overfitting and improve the model's generalization ability, we set the dropout rate to 0.5. The specific hyperparameter values are shown in Table 2.

For a fair comparison, during the evaluation phase, we followed previous research [18,35] and adopted bootstrapping sampling rather than cross-validation. Specifically, we randomly sampled 80% of the data from the test set for evaluation and repeated this process 10 times. The average results of these runs are reported as the final outcome. All benchmark models compared in this study relied on the optimal results provided by the authors in the referenced literature.

5.4. Analysis of experimental results (RQ1)

Table 3 presents the performance comparison of various models on the MIMIC-III dataset. As shown in the table, the longitudinal-based models outperform the three instance-based methods (LR, ECC, and LEAP). This is primarily because instance-based methods rely solely on the diagnosis and records from the patient's current medical visit, ignoring their past medical history. RETAIN uses RNNs to encode longitudinal patient history; GAMENet utilizes graph convolutional networks to integrate patient information with DDI information, improving medication recommendation performance; SafeDrug is the first to consider the molecular structure of medications in the recommendation process, using DDI at the molecular level to reduce the DDI rate; COGNet introduces a replication/prediction mechanism to generate medication sets, using historical recommendations to supplement current medication recommendations for more accurate predictions; MoleRec effectively identifies relationships between substructure perceptions by modeling interactions between medication substructures. StartMed solves the long-tail problem in current EHR data using a stratification strategy, enabling comprehensive learning of relatively sparse data. ACDNet uses attention mechanisms and Transformer models to capture the patient's health status and medication history by modeling medical records globally and locally. FDIRNet adopts a forward data interpolation method to handle missing and anomalous data from both single and multiple visits, effectively increasing the amount of usable data.

Compared to the baseline methods, SDRBT introduces a deep spatio-temporal encoding module that combines longitudinal symptom information, diagnosed disease information, treatment information, and related multi-dimensional data from patient EHRs. By using the Block Recurrent Transformer instead of the traditional Transformer to aggregate longitudinal feature information, SDRBT effectively resolves the issue of loosely connected longitudinal patient information. Additionally, SDRBT incorporates a medication substructure dual-domain mapping module, which introduces an attention learning method based on medication molecular substructures. Through global and local mapping of medication molecular substructures, the model captures specific substructures that play a significant role in treating patients' diseases, providing more accurate medication combinations. Furthermore, considering the interactions between medications, we integrate the EHR graph with the DDI graph and perform similarity calculations. This allows us to identify key medications used in the patient's treatment process and supplement missing drug interactions in the DDI graph. This approach not only enhances the safety of the medication package but also accelerates the model's training speed. Finally, we incorporate the similarity-calculated DDI control module into our designed PID Loss control unit. By intervening in the probability of each medication appearing in the recommended medication package during training, we ensure the safety of the final recommended medication combination. Overall, SDRBT outperforms all baselines in terms of Jaccard, F1-score, and PRAUC, demonstrating that our method effectively improves recommendation efficiency while ensuring the safety of the recommended medication combinations. This experiment addresses RQ1.

5.5. Ablation experiment (RQ2)

In this section, we designed five simplified variants of the model to test the impact of each model component on the model performance. The relevant experimental results are shown in Table 4. This experiment solves RQ2.

- The w/o sym. variant removes patient symptom information in the model multi-dimensional information aggregation module.
- The w/o BRT replaces the Block Recurrent Transformer module in the longitudinal characterization aggregation module with the common Transformer encoding.
- The w/o SC variant removes the similarity between the EHR diagram and the DDI diagram in the safe medication combination recommendation module.
- The w/o Lddi variant removes the DDI control module in the PID Loss control unit.
- The w/o DDM variant removes the model medication substructure dual domain mapping module (DDM).

By analyzing the results in Table 4, we can conclude that in the ablation experiment of the SDRBT model: (1) The performance of the w/o sym. variant decreased by approximately 3.54%, indicating that the model achieved a full integration of different dimensions of patient information by incorporating patient symptom information. (2) The w/o BRT variant experiments have shown that EHR timing data information of patients is indeed not compact, and Transformer is worse than Block Recurrent Transformer in processing such timing information. (3) After the process of finding the similarity between EHR plot and DDI plot was removed from the model, the performance of the model was decreased through the ablation data of the w/o SC variant. This shows that the process can effectively screen out the medications used in the treatment of patients from the DDI map and supplement the medications that do not appear. (4) The DDI value of the w/o Lddi variant increased by about 39.26% after the DDI control module in the PID Loss control unit was deleted, and the safety of the recommended medication combination decreased. This indicates that the PID Loss control unit designed by us can effectively reduce the interaction between recommended medications while ensuring the recommendation effect, and improve the safety of the final recommended medication package. (5) Compared with other modules, all indexes of the w/o DDM variant decreased to a greater extent after removing the medication substructure dual domain mapping module, and the main index Jaccard decreased by about 6.19%. This indicates that the module can well model the global and local medication molecular substructure, and obtain the contribution of different medication molecular structures to the cure of patients' diseases. Overall, SDRBT outperformed all model variants, meaning that each component of our model contributed to the overall performance improvement.

Table 3
Comparison of model performance on MIMIC-III data set.

Model	Jaccard	F1-score	PRAUC	DDI
LR	0.4865 ± 0.0021	0.6434 ± 0.0019	0.7509 ± 0.0018	0.0829 ± 0.0009
ECC(2011)	0.4996 ± 0.0049	0.6569 ± 0.0044	0.6844 ± 0.0038	0.0846 ± 0.0018
LEAP(2017)	0.4521 ± 0.0024	0.6138 ± 0.0026	0.6549 ± 0.0033	0.0731 ± 0.0008
RETAIN(2016)	0.4887 ± 0.0028	0.6481 ± 0.0027	0.7556 ± 0.0033	0.0835 ± 0.0020
GAMENet(2019)	0.5067 ± 0.0025	0.6626 ± 0.0025	0.7631 ± 0.0030	0.0864 ± 0.0006
SafeDrug(2021)	0.5213 ± 0.0030	0.6768 ± 0.0027	0.7647 ± 0.0025	**0.0589 ± 0.0005**
COGNet(2022)	0.5336 ± 0.0011	0.6869 ± 0.0010	0.7739 ± 0.0009	0.0852 ± 0.0005
MoleRec(2023)	0.5305 ± 0.0033	0.6843 ± 0.0029	0.7736 ± 0.0027	0.0724 ± 0.0008
StartMed(2024)	0.5321 ± 0.0035	0.6861 ± 0.0034	0.7779 ± 0.0043	0.0642 ± 0.0005
ACDNet(2024)	0.5433 ± 0.0027	0.6957 ± 0.0021	0.7904 ± 0.0021	0.0859 ± 0.0010
FDIRNet(2024)	0.5502 ± 0.0010	0.6982 ± 0.0011	0.7983 ± 0.0008	0.0691 ± 0.0006
SDRBT(ours)	**0.5592 ± 0.0021**	**0.7097 ± 0.0016**	**0.8011 ± 0.0014**	0.0624 ± 0.0009

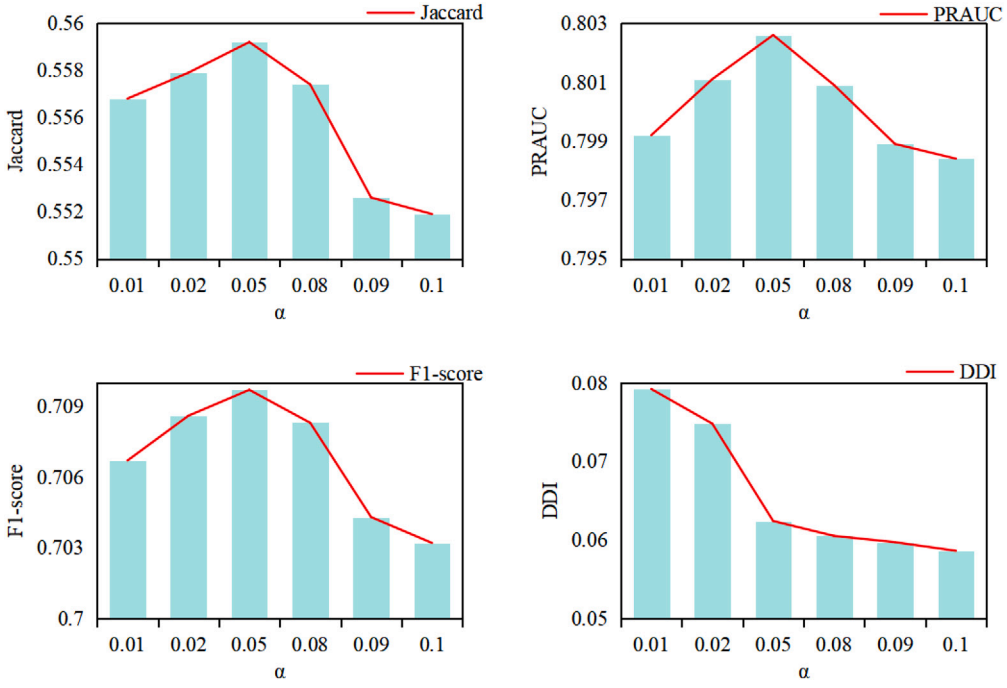


Fig. 3. The indexes of different models vary with Lddi weight α.

Table 4
Ablation of SDRBT model on MIMIC-III dataset.

Model	Jaccard	F1-score	PRAUC	DDI
SDRBT(w/o sym.)	0.5401	0.6931	0.7878	0.0683
SDRBT(w/o BRT)	0.5423	0.6941	0.7858	0.0698
SDRBT(w/o SC)	0.5582	0.7091	0.8006	0.0682
SDRBT(w/o Lddi)	0.5553	0.7057	0.7986	0.0869
SDRBT(w/o DDM)	0.5256	0.6782	0.7752	0.0736
SDRBT	**0.5592**	**0.7097**	**0.8011**	**0.0624**

Table 5
Comparison of experimental results of each index of the model with Lddi weight α.

Model	Jaccard	F1-score	PRAUC	DDI
SDRBT(α_{ddi} = 0.1)	0.5519	0.7032	0.7984	0.0586
SDRBTα_{ddi} = 0.09)	0.5526	0.7043	0.7989	0.0597
SDRBT(α_{ddi} = 0.08)	0.5574	0.7083	0.8009	0.0605
SDRBTα_{ddi} = 0.05	**0.5592**	**0.7097**	**0.8026**	**0.0624**
SDRBT(α_{ddi} = 0.02)	0.5579	0.7086	0.8011	0.0748
SDRBT(α_{ddi} = 0.01)	0.5568	0.7067	0.7992	0.0792

In addition, in order to achieve the ideal trade-off of DDI rate for SDRBT in the recommended medication combination, we explored the changes of each evaluation index of the model with different Lddi weight α, and the experimental results were shown in Table 5 and Fig. 3. According to the experimental results, we can get: (1) With the increase of weight α, the DDI rate between the medication combinations predicted by the model gradually decreases. (2) When $\alpha < 0.05$, the recommendation performance of the model is improved with the increase of weight α; When $\alpha > 0.05$, the performance of the model fluctuates with the increase of weight α, and the indexes show a downward trend. The experimental results show that there is an equilibrium point in the adjustment process of Lddi weight α, so we need to comprehensively consider the relationship between DDI rate and model prediction accuracy. In this paper, SDRBT selected $\alpha = 0.05$ as the weight of the Loss function used by Lddi in the final training process.

5.6. Efficiency analysis (RQ3)

In this section, we conducted an efficiency experiment to validate the time efficiency of different models when trained on the same dataset. All experiments were performed on the same device, with the computation method involving recording the total time taken by the model to complete the same number of epochs, and then dividing the total time by the total number of epochs to obtain the time taken for one epoch.

Table 6
Runtime comparison between different models (in seconds).

Model	First	Second	Third	Fourth	Fifth	Average time
SafeDrug	147.73 s/epoch	154.72 s/epoch	137.17 s/epoch	154.58 s/epoch	147.60 s/epoch	148.36 s/epoch
MoleRec	449.76 s/epoch	438.56 s/epoch	435.83 s/epoch	451.5 s/epoch	425.20 s/epoch	440.17 s/epoch
COGNet	613.01 s/epoch	637.75 s/epoch	614.59 s/epoch	621.05 s/epoch	604.90 s/epoch	618.26 s/epoch
SDRBT	82.76 s/epoch	79.05 s/epoch	83.37 s/epoch	82.26 s/epoch	77.36 s/epoch	80.96 s/epoch

Table 7
Example of recommended medications by SDRBT for a given patient's health condition.

Case	Diagnosis	Correct	Unseen	Missed
1	03849, 5770, 5849, 5761, 1623, 5859, 496, 99592, 57450, 7904, 28800, V1582, 40390, 4439, 2720, 28522, 1991	13: D06A, J01C, N05A, J01M, A12A, NO2A, A01A, B05C, NO2B, B01A, A12C, A02B, A06A	2	3
2	241, 416, 248, 83, 484, 1036, 239, 510, 46, 4, 78, 1939	14: N07A, C07A, B03B, A12C, N05A, A06A, A04A, A10A, B05C, N05C, N02B, N03A, N03C, C02D	3	4
3	03842, 78552, V427, 99592, 4019, 25000, V1007	16: J05A, D01A, J01E, R05C, A07E, NO5B, J01D, N06A, A12A, NO2A, A12B, A01A, NO2B, A12C, A02B, A06A	3	5

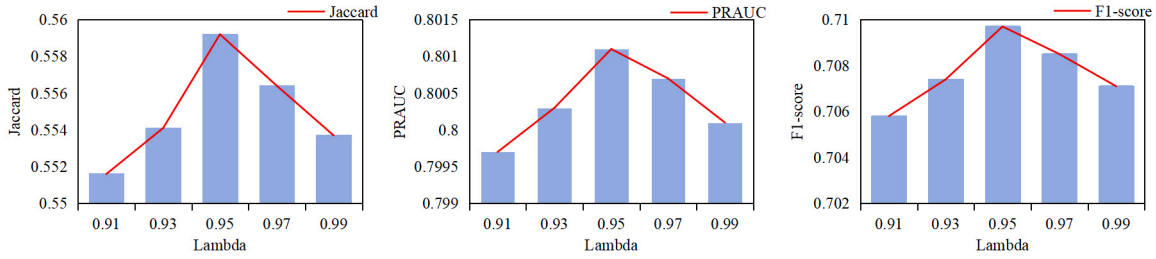


Fig. 4. The impact of parameter λ on model performance on the MIMIC-III dataset.

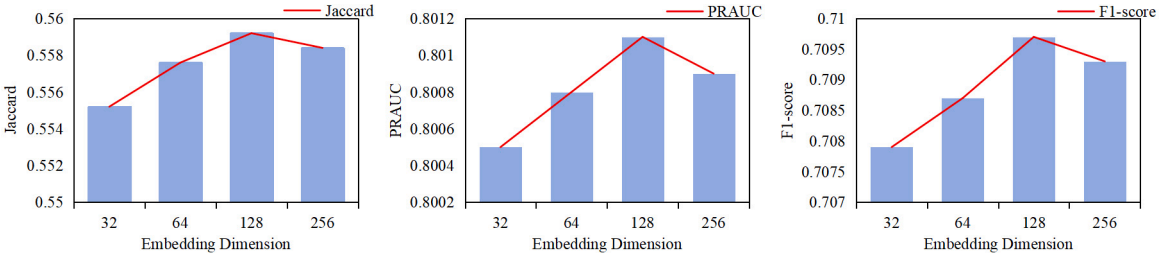


Fig. 5. The impact of embedding dimensions on model performance on the MIMIC-III dataset.

We selected SafeDrug, MoleRec and COGNet models with relatively good time efficiency among the baseline models for efficiency comparison with our model SDRBT. Table 6 shows the results of the time efficiency comparison. Upon comparing the average training time per epoch, SDRBT demonstrates a significantly shorter training time compared to the baseline models. This signifies that our model achieves a balance between maintaining good time efficiency while also delivering high accuracy and precision. Furthermore, it validates that SDRBT, by integrating the EHR graph with the DDI graph and performing similarity calculations, can effectively screen for key medications used in patient treatment, thereby reducing training time and enhancing model efficiency. This experiment addresses RQ3.

5.7. Case study

To intuitively demonstrate the performance of the SDRBT model, we analyzed three specific examples and presented the model's prediction results. In the validation process, we used the given patient's diagnostic information and the trained SDRBT model to recommend medication combinations for the patient. The example results are shown in Table 7, where the diagnostic information is represented using the International Classification of Diseases (ICD) codes and the medications are represented using ATC codes. In the table, "Correct" indicates the medications correctly predicted by the model, "Missed" represents the medications that are in the true medication set but were not predicted by the model, and "Unseen" refers to the medications predicted by the model that are not in the true medication set.

Overall, SDRBT exhibited superior performance in each medication recommendation, being able to not only accurately predict the medication set but also effectively minimize the total number of recommended medications. This enables the model to ensure the accuracy of medication recommendations while significantly improving the efficiency of the recommendations, reducing redundancy in the recommended medications, and thus providing more precise and efficient medication combination recommendations.

5.8. Parameter impact

In this section, we evaluate the performance of the SDRBT model on the MIMIC-III dataset using various parameter settings. Specifically, we explore different combinations of λ values and embedding dimensions to identify the optimal configuration that yields the best performance.

Firstly, with the embedding dimension fixed at 128, we examine the impact of different λ values on model performance. As shown in Fig. 4,

the Jaccard, PR-AUC, and F1 scores reach their peak when λ is set to 0.95.

Subsequently, with a fixed λ value of 0.95, we further analyze the effect of different embedding dimensions on model performance. The results shown in Fig. 5 indicate that an embedding dimension of 128 achieves the highest Jaccard, PR-AUC, and F1 scores.

6. Conclusion

In this study, we propose a medication recommendation model called "SDRBT". This research adopts the Block Recurrent Transformer to replace the traditional Transformer for modeling patients' longitudinal temporal information, breaking the previous modeling approach for patient EHR data. By combining patients' symptom information with multi-dimensional data, it addresses the potential limitations of patients' current visit representations. Additionally, we introduce a medication substructure dual-domain mapping module, where medications undergo global and local dual structural mapping. This effectively learns and aggregates medication substructure representations, obtaining a perceptual representation of medication substructures that play a critical role in curing patients' diseases. Finally, the model integrates the EHR graph with the DDI graph and performs similarity calculations to obtain the DDI control module. This reduces the potential harm caused by different medication combinations to the human body during the training process, ensuring the safety of medication combinations. Experimental results on the public MIMIC-III dataset demonstrate the superiority of SDRBT compared to existing medication recommendation models. Further ablation study results also indicate the effectiveness of each module in the model.

CRediT authorship contribution statement

Haoqin Yang: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Yuandong Liu:** Conceptualization. **Longbo Zhang:** Project administration, Conceptualization. **Hongzhen Cai:** Project administration, Conceptualization. **Kai Che:** Project administration, Conceptualization. **Linlin Xing:** Supervision, Project administration, Investigation, Data curation, Conceptualization.

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Code availability

Our source code of SDRBT is available at https://github.com/yhqfighting/SDRBT.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Linlin Xing reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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